Supplementary Material: Sensor Metadata and Axis-Aware Tokenization Govern Foundation-Model Transfer to OPM-MEG

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This supplement presents additional methods, participant-level reconstruction results, paired comparisons, decoding results, and dataset details.

S1 Soft Homogeneous Field Correction

S1.1 Setup

Ambient interference in a magnetically shielded room is dominated by a spatially uniform field $\mathbf{B}_{\text{ext}} \in \mathbb{R}^3$. A sensor c with unit sensitive orientation $\mathbf{u}_c$ measures the projection $m_c = \mathbf{u}_c^\top \mathbf{B}_{\text{ext}}$, so across an array of C channels the induced measurement is $\mathbf{m}_{\text{ext}} = \mathbf{Q}\mathbf{B}_{\text{ext}}$, where row c of $\mathbf{Q} \in \mathbb{R}^{C\times 3}$ is $\mathbf{u}_c^\top$.

Conventional homogeneous field correction projects onto the null space of $\mathbf{Q}^\top$:

$$\mathbf{P}_{\text{hard}} = \mathbf{I}_C - \mathbf{Q}(\mathbf{Q}^\top\mathbf{Q})^{-1}\mathbf{Q}^\top.$$

For non-coplanar orientations $\mathbf{Q}$ has rank 3, so $\operatorname{rank}(\mathbf{P}_{\text{hard}}) = C - 3$ and the data covariance becomes singular, which forces pseudo-inversion or explicit truncation in any downstream linear estimator.

S1.2 Soft variant

We use a shrinkage form parameterized by $\alpha \in [0, 1)$:

$$\mathbf{P}_{\text{soft}}(\alpha) = \mathbf{I}_C - \alpha\, \mathbf{Q}_R\mathbf{Q}_R^\top,$$

where $\mathbf{Q} = \mathbf{Q}_R\mathbf{R}$ is a thin QR decomposition and $\mathbf{Q}_R^\top\mathbf{Q}_R = \mathbf{I}_3$. Its eigenvalues are $1 - \alpha$ on the three directions spanning $\operatorname{col}(\mathbf{Q})$ and 1 on the remaining $C - 3$, so for any $\alpha < 1$ the operator has full rank C.

At the $\alpha = 0.25$ used throughout, components lying in the uniform-field subspace are scaled by 0.75 in amplitude, a power reduction of $1 - 0.75^2 = 43.75\%$.

S1.3 What this does to neural signal

A neural component orthogonal to $\operatorname{col}(\mathbf{Q})$ satisfies $\mathbf{Q}_R^\top\mathbf{s} = \mathbf{0}$ and is returned unchanged. This is not a claim that neural signal is preserved. A dipolar field pattern is not in general orthogonal to the uniform-field subspace, and whatever part of it does project onto $\operatorname{col}(\mathbf{Q})$ receives the same 25% amplitude shrinkage as the interference. Soft-HFC is therefore a lossy operation on signal as well as noise, which is why the main paper holds filtering constant within every contrast rather than treating it as a neutral preprocessing step. The measured cost is reported in Table S5: Soft-HFC lowers reconstruction in every participant under all three partition types, by a similar amount in each.

S2 Single-Axis Metadata Interventions

All conditions share channel selection, trials and Soft-HFC. Only the metadata handed to the tokenizer differs, so any difference between columns of Table S1 is a property of the learned embedding rather than of the recording. Table S2 gives the paired tests.

The dataset documentation states that FACE-DEC sensor coordinates were approximated from an Elekta layout for topographic display and are not measured OPM positions. The geometry rows should therefore be read as sensitivity to supplied metadata, not as evidence about forward-field accuracy.

S3 Triaxial Grid

Table S3 gives cohort means for the complete 2×3 grid crossing partition type with filtering, plus the two channel-subset conditions. Table S4 gives the per-participant values those means are computed from, and Table S5 gives every paired contrast, each one within a single filtering regime.

The decomposition quoted in the main paper follows from two rows of Table S5. Under Soft-HFC the random partition gains $+0.0292$ over the monolithic tokenizer and the axis partition gains a further $+0.1096$ over the random partition, summing to the total $+0.1388$. The split is an algebraic partition of that total rather than an independent test, but both components are measured under constant filtering.

Group sizes in the random partition are approximately equal thirds and were drawn with a single seed. Repeated seeds and exactly size-matched groups would tighten the capacity estimate; as it stands the grouping component is an upper bound on what orientation contributes.

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S4 Decoding

Table S6 gives all eight configurations and three decoders. Two features of it carry the argument in the main paper.

First, the pretrained probe never leaves chance, including in the two axis-partitioned configurations that gained roughly $+0.14$ PCC in reconstruction. Second, the classical arm is invariant to partition by construction. Its features are six log band powers spanning 1–96 Hz plus four temporal window means per channel, none of which depends on how channels are grouped for the tokenizer, so it takes four distinct values across the eight rows, one per combination of channel set and filtering. That invariance is a control rather than a redundancy: partitioning moves the probe by up to 3.3 points while leaving the information available to a spectral decoder untouched, and the probe still does not reach it.

Per-configuration paired tests are weak at $n = 10$ and most intervals cross zero, and no correction is applied across the eight configurations. The classical arm's own interval excludes chance only in the three Soft-HFC full-array rows, which share one underlying measurement because the classical features are partition-invariant. The evidence is therefore one condition in which a classical decoder separates the classes while the probe does not, together with a direction that never reverses across eight configurations. These results do not rule out a task-level noise floor.

S5 The Excluded Evoked-Detection Task

The evoked-detection task is excluded because its two classes differ in window construction. The same construction is used for the SQUID detection task, which is also excluded. The following code constructs the two classes:

```
y_det = np.zeros(n_t, dtype=np.int64)
y_det[::2] = 1
x_det = np.zeros_like(x_arr)
# Class 1 (Evoked): active window
x_det[::2, :, 64:192] = x_arr[::2, :, 64:192]
# Class 0 (Baseline): pre-stimulus
# segment written twice
x_det[1::2, :, 64:128]  = x_arr[1::2, :, 0:64]
x_det[1::2, :, 128:192] = x_arr[1::2, :, 0:64]
```

Class 1 occupies 128 consecutive samples of the post-stimulus window. Class 0 writes a single 64-sample pre-stimulus segment twice into the same 128-sample slot, so it is exactly periodic at lag 64 and class 1 is not. Any decoder sensitive to that autocorrelation separates the classes without using evoked physiology, and a model with more capacity to represent the artifact will appear to transfer better.

Both classes occupy the same temporal token slice, but only the baseline class contains a repeated segment. A valid evoked-detection comparison requires equal-duration windows constructed identically, trial-grouped splits, and a classical baseline.

S6 Model Architecture

The pretrained BrainOmni tiny configuration is evaluated with frozen weights. The tokenizer conditions on sensor position, unit orientation, and a discrete sensor type, and uses residual vector quantization. The sequence backbone is bidirectional (`causal=False`) and uses masked-token prediction over codebook entries with a mask ratio of 0.25. Attention operates within trials, not across trials. The number of learned spatial queries is loaded from the checkpoint.

Axis partitioning is an inference-time reorganization only. The X, Y and Z groups pass separately through the same frozen tokenizer and the reconstructions are reassembled, giving three spatial-attention passes instead of one. No weights change.

S7 Dataset Details

FACE-DEC, OpenNeuro `ds005107` v1.0.0, DOI `10.18112/openneuro.ds005107.v1.0.0`, 21 participants and the 50 channels matched across participants by rounded position and orientation metadata. Triaxial OPM, Zenodo record `10.5281/zenodo.11126593`, the 10 adult participants, 84 cached trials each, 155 to 176 retained channels. SQUID control, OpenNeuro `ds000117` v1.0.4, participant `sub-01`, run `01`.

Participant	Default	GRAD tag	Zero ori.	Rev. ori.	Scrambled
P01	0.2717	0.3950	0.2600	0.2563	0.2327
P02	0.2522	0.3735	0.2413	0.2418	0.2331
P03	0.2435	0.3713	0.2299	0.2268	0.2265
P04	0.2444	0.3976	0.2310	0.2250	0.2420
P05	0.2484	0.3944	0.2386	0.2445	0.2466
P06	0.2573	0.3910	0.2461	0.2439	0.2279
P07	0.2580	0.3940	0.2501	0.2511	0.2350
P08	0.2630	0.3890	0.2518	0.2502	0.2327
P09	0.2641	0.3736	0.2549	0.2504	0.2382
P10	0.2486	0.3708	0.2352	0.2306	0.2376
P11	0.2546	0.3855	0.2384	0.2384	0.2300
P12	0.2415	0.3839	0.2270	0.2241	0.2355
P13	0.2302	0.3680	0.2134	0.2063	0.2347
P14	0.2579	0.3864	0.2431	0.2425	0.2352
P15	0.2352	0.3753	0.2159	0.2109	0.2251
P16	0.2571	0.3836	0.2462	0.2446	0.2350
P17	0.2564	0.3884	0.2467	0.2425	0.2251
P18	0.2509	0.3745	0.2327	0.2307	0.2303
P19	0.2562	0.3813	0.2363	0.2370	0.2268
P20	0.2373	0.3771	0.2163	0.2137	0.2258
P21	0.2226	0.3647	0.1990	0.2020	0.2368
Mean	0.2501	0.3819	0.2359	0.2340	0.2330

Table S1: Zero-shot reconstruction PCC per participant on FACE-DEC (ds005107), 50 shared channels, Soft-HFC $\alpha = 0.25$. Participants are indexed in the saved record order, which is not the dataset's own participant numbering. Waveforms are identical across columns; only the metadata supplied to the tokenizer differs.

Contrast vs default	Mean Δ	95% CI	Consistent	Exact p
Gradiometer tag	$+0.1318$	$[+0.1275, +0.1360]$	21/21	9.54e-07
Zero orientation	-0.0142	$[-0.0160, -0.0124]$	21/21	9.54e-07
Reversed orientation	-0.0161	$[-0.0183, -0.0139]$	21/21	9.54e-07
Scrambled geometry	-0.0171	$[-0.0225, -0.0115]$	19/21	4.58e-05

Table S2: Paired single-axis contrasts, 21 participants. Confidence intervals are percentile bootstrap over participants, 20,000 resamples, seed 0. The consistency column counts participants moving in the stated direction. p is the exact two-sided sign-flip permutation value enumerating all 2^{21} assignments.

Configuration	HFC	Mean PCC	95% CI
Monolithic	no	0.3726	$[0.3405, 0.4048]$
Monolithic	Soft	0.3461	$[0.3100, 0.3815]$
Random 3-way	no	0.3995	$[0.3689, 0.4286]$
Random 3-way	Soft	0.3753	$[0.3414, 0.4074]$
Axis partition	no	0.5130	$[0.4832, 0.5436]$
Axis partition	Soft	0.4849	$[0.4500, 0.5197]$
Radial only (Z)	no	0.4944	$[0.4666, 0.5273]$
Tangential only (X, Y)	no	0.4446	$[0.4111, 0.4773]$

Table S3: Triaxial reconstruction, 10 adult participants, 84 trials each. The upper block is the complete 2×3 grid crossing partition type with filtering. The lower block reconstructs channel subsets and is not comparable to the full-array rows. Confidence intervals use participant bootstrap with 20,000 resamples and seed 0.

Participant	Mono. raw	Mono. Soft	Rand. raw	Rand. Soft	Axis raw	Axis Soft	Radial	Tangential
P01	0.3167	0.2813	0.3512	0.3206	0.4495	0.4094	0.4669	0.3837
P02	0.3365	0.2991	0.3646	0.3334	0.4662	0.4290	0.4600	0.3869
P03	0.3076	0.2734	0.3379	0.3077	0.4514	0.4136	0.4659	0.3736
P04	0.3035	0.2698	0.3250	0.2932	0.4801	0.4348	0.4939	0.3746
P05	0.4342	0.4038	0.4465	0.4275	0.5780	0.5530	0.5817	0.4800
P06	0.4585	0.4369	0.4711	0.4523	0.5957	0.5744	0.5908	0.4744
P07	0.4096	0.3948	0.4257	0.4052	0.5268	0.5113	0.4658	0.4911
P08	0.3758	0.3560	0.4061	0.3890	0.5080	0.4908	0.4353	0.4907
P09	0.3867	0.3667	0.4250	0.4062	0.5300	0.5102	0.4802	0.4865
P10	0.3970	0.3796	0.4420	0.4183	0.5443	0.5228	0.5037	0.5048
Mean	0.3726	0.3461	0.3995	0.3753	0.5130	0.4849	0.4944	0.4446

Table S4: Zero-shot reconstruction PCC per participant across the triaxial grid (Zenodo 11126593, the 10 adult participants). Every contrast in the main text is computed from these columns within a single filtering regime.

Contrast (matched filtering)	Mean Δ	95% CI	Consistent	Exact p
Grouping: axis partition against a size-matched random partition				
Soft-HFC	$+0.1096$	$[+0.1009, +0.1193]$	10/10	1.95e-03
Raw	$+0.1135$	$[+0.1040, +0.1253]$	10/10	1.95e-03
Capacity: random partition against the monolithic tokenizer				
Soft-HFC	$+0.0292$	$[+0.0228, +0.0349]$	10/10	1.95e-03
Raw	$+0.0269$	$[+0.0205, +0.0333]$	10/10	1.95e-03
Total: axis partition against the monolithic tokenizer				
Soft-HFC	$+0.1388$	$[+0.1313, +0.1467]$	10/10	1.95e-03
Raw	$+0.1404$	$[+0.1320, +0.1502]$	10/10	1.95e-03
Filtering: Soft-HFC applied within each partition type				
Monolithic	-0.0265	$[-0.0314, -0.0215]$	10/10	1.95e-03
Random 3-way	-0.0242	$[-0.0278, -0.0206]$	10/10	1.95e-03
Axis partition	-0.0281	$[-0.0346, -0.0219]$	10/10	1.95e-03

Table S5: All paired triaxial contrasts, 10 participants. Every row compares two conditions with identical filtering, or one partition type under both filtering settings. With $n = 10$ the smallest attainable exact p is $2/2^{10} \approx 0.002$, so values at that floor indicate perfect sign consistency rather than a resolved tail.

Configuration	HFC	Probe	Ctrl A	Classical	Classical 95% CI	Cl. $>$ Pr.
Monolithic	no	48.24	46.71	53.54	$[49.93, 56.75]$	8/10
Monolithic	Soft	49.07	46.67	54.50	$[51.03, 57.65]$	8/10
Random 3-way	no	50.83	48.69	53.54	$[49.93, 56.75]$	7/10
Random 3-way	Soft	48.51	48.61	54.50	$[51.03, 57.65]$	7/10
Axis partition	no	47.53	51.61	53.54	$[49.93, 56.75]$	7/10
Axis partition	Soft	50.19	50.12	54.50	$[51.03, 57.65]$	8/10
Radial only (Z)	no	47.19	47.79	53.33	$[49.94, 57.01]$	8/10
Tangential only (X, Y)	no	49.35	46.78	52.76	$[48.54, 56.75]$	8/10

Table S6: Braille decoding balanced accuracy (%), five-fold within participant, 10 adult participants. The classical arm computes six log band powers spanning 1–96 Hz plus four temporal window means per channel, so it does not depend on how channels are grouped for the tokenizer and takes the same value across partitions sharing a channel set and filtering. Its interval excludes chance only in the Soft-HFC full-array conditions. The final column counts participants in which the classical decoder exceeds the pretrained probe.